

EXAMPREP AI

1. KEY PERFORMANCE INDICATORS (KPIs)

Overview

The KPIs will be used to measure the success of the ExamPrep AI MVP. Each KPI aligns with the product objectives and success indicators and will be tracked using application events collected through the analytics platform.

- **North Star Metric**

Name: Recommended Topic Completion Rate

Definition: The percentage of active students who complete at least one topic recommended by the study planner.

Purpose: It directly measures whether students are acting on the product's core value proposition—using personalized recommendations to guide what they study next.

Formula: $(\text{No of active students who complete at least One Recommended Topic}) \div (\text{Total number of active students}) \times 100$

- **Supporting KPIs**

KPIs	Definition	Formula	Purpose
Activation Rate	Percentage of new users who complete their first quiz within 24 hours of registering.	$(\text{Sign-ups completing 1st quiz within 24h}) \div (\text{Total new sign-ups}) \times 100$	Indicates how effectively the application converts new users into active learners
Quiz Completion Rate	The percentage of quizzes that are completed after being started.	$(\text{Completed Quizzes} \div \text{Started Quizzes}) \times 100$	Measures whether students successfully complete the application's core learning activity.
7-Day Retention Rate	percentage of new users who return and remain active seven days after registration.	$(\text{Users active by day 7}) \div (\text{Total new users}) \times 100$	Measures whether the product provides enough value for students to continue using it beyond their initial session.
Topic Accuracy Improvement	Average change in a student's accuracy on a topic after completing the recommended revision for that topic	$\Sigma (\text{average post-revision topic accuracy} - \text{average pre-revision topic accuracy}) \div \text{Total number of revised topics}$	Measures whether personalized study recommendations are contributing to improved understanding of previously weaker topics

2. USER DATA REQUIREMENTS

Data Field	Purpose
Full Name	Identifies the student within the application.
Email Address	Used for account registration, login, and account recovery.
User ID	Uniquely identifies each user throughout the application and database.
Target Examination (JAMB/WAEC)	Personalizes quizzes, study plans, and recommendations based on the user's exam.
Preferred Subject	Determines the subject-specific quiz questions and study recommendations presented to the user.
Registration Date & Time	Records when the account was created for activation and retention analysis.
Last Active Date & Time	Tracks the user's most recent activity for engagement, retention, and study streak calculations.

LEARNING & ACTIVITY DATA REQUIREMENTS

Data Field	Purpose
Quiz ID	Identify each quiz attempt.
Subject ID	Determine the subject attempted.
Topic ID	Track topic performance.
Question ID	Track responses to individual questions.
Quiz Score	Measure quiz performance.
Topic Accuracy (%)	Identify weak and strong topics.
Recommendation Status	Measure whether recommendations are acted upon.
Study Streak Count	Measure learning consistency.
Quiz Start Time	Calculate Quiz Completion Rate.
Quiz Completion Time	Measure completion and duration.

3. ANALYTICS EVENT TRACKING PLAN

The Event Tracking Plan defines the user actions and system events that must be captured during interaction with ExamPrep AI. Collecting these events ensures that the data required to calculate the North Star Metric, supporting KPIs, and other product analytics is available for analysis.

Event Name	Trigger	Key Properties
sign_up_completed	User successfully completes account registration.	user_id, signup_method, timestamp
subject_selected	User selects a subject to study.	user_id, subject_id, timestamp
quiz_started	User begins a quiz.	user_id, quiz_id, subject_id, topic_id, question_count, is_recommended, timestamp
question_answered	User submits an answer to a quiz question.	user_id, quiz_id, question_id, topic_id, is_correct, time_taken_sec, timestamp
quiz_completed	User successfully completes and submits a quiz.	user_id, quiz_id, subject_id, topic_id, score, accuracy, is_recommended, completion_time_sec, timestamp
performance_viewed	User opens the performance summary page.	user_id, quiz_id, timestamp
study_plan_viewed	User opens the personalized study plan.	user_id, quiz_id, recommended_topics, timestamp
recommended_topic_revised	User completes revision for a topic recommended by the study planner.	user_id, subject_id, topic_id, timestamp
streak_incremented	User's study streak increases after an eligible study activity.	user_id, streak_count, timestamp
streak_reset	User's study streak is broken and reset.	user_id, previous_streak, timestamp

4. DATA FLOW DIAGRAM

